

A Self-Closing Gate Conundrum

#1 Victory Royale

Element I: Testing, Data Collection and Analysis

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

The cost of all the materials needed for our final prototype is \$41.04. We will also have extra of our nuts, bolts, and galvanized hanger strap. The cost of our prototype that we made in class is a no-cost prototype because we used materials found inside the class.

Our testing procedures consist of opening the gate at different forces and seeing the force at which it closes. We will test that four times and record data. The spring and clamp have to close a gate that is 20 lbs.



	Test 1	Test 2	Test 3	Test 4	Average
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Open gate normally	The gate opened perfectly when opened normally.	The gate opened perfectly when opened normally.	The gate opened perfectly when opened normally.	The gate opened perfectly when opened normally.	The gate opened perfectly when opened normally.
Open gate forcefully	The gate opened perfectly when opened forcefully.	The gate opened perfectly when opened forcefully.	The gate opened perfectly when opened forcefully.	The gate opened perfectly when opened forcefully.	The gate opened perfectly when opened forcefully.
Measurements	2 feet length, $\frac{3}{4}$ inch width and height				
Weight	146.0 grams				

